APPLIED STATISTICS, PROBABILITY THEORY AND MATHEMATICAL STATISTICS

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Nonparametric statistics

Exercise 1.2

Check is a die is fair or not. Let be A_i the event, that we rolled i with the die ($i = 1, \ldots, 6$). Now $H_0 : P(A_i) = \frac{1}{6},\ i = 1, \ldots, 6$.

We rolled the die 600 times and got the following result:

$$k_1 = 83, \qquad k_2 = 91, \qquad k_3 = 122, \qquad k_4 = 107, \qquad k_5 = 74, \qquad k_6 = 123$$

(k_i is the number of cases when the outcome was A_i).

$$\chi^2 = \frac{(83 - 100)^2}{100} + \frac{(91 - 100)^2}{100} + \frac{(122 - 100)^2}{100} + \\ + \frac{(107 - 100)^2}{100} + \frac{(74 - 100)^2}{100} + \frac{(123 - 100)^2}{100} = 21.08.$$

If H_0 is true then χ^2 statistics asymptotically χ^2_5 distributed. We reject H_0 on $1 - \alpha$ degree, if χ^2 exceeds the critical value of the $1 - \alpha$ degree. You can find the critical value in the table of distribution of χ^2_5. If $\alpha = 0.1$ the value is 9.236, if $\alpha = 0.01$ the value is 15.09, if $\alpha = 0.001$ the value is 20.51. In conclusion we reject H_0 on every degree.

Exercise 1.2

```
die = 1:6;
 freq = [83,91,122,107,74,123];
 expected = repmat(100, 1,6) % [100,100,100,100,100,100]
 [h,p,stats] = chi2gof(die,"Expected",expected,"Frequency",freq)
```

Exercise 1.5

Test if the distribution of the random variable X is a Poisson distribution with $\lambda = 2$ parameter. Let's use the following partition from the Poisson distribution table: $A_i = \{X = i\},\ i = 0, 1, 2, 3$ and $A_4 = \{X \geq 4\}$

$P_{H_0}(A_0) = 0.135,\quad P_{H_0}(A_1) = 0.270,\quad P_{H_0}(A_2) = 0.270,\quad P_{H_0}(A_3) = 0.180,$
$P_{H_0}(A_4) = 0.145.$

Taking $N = 100$ samples for X, the frequency of event A_k the following respectively

$$12, 32, 25, 21, 10$$

Now

$$\chi^2 = \frac{(12 - 13.5)^2}{13.5} + \frac{(32 - 27)^2}{27} + \frac{(25 - 27)^2}{27} + \frac{(21 - 18)^2}{18} + \frac{(10 - 14.5)^2}{14.5} =$$
$$= 0.166 + 0.926 + 0.148 + 0.5 + 1.396 = 3.136.$$

On $1 - \alpha = 0.9$ degree from the χ_4^2 table we get the 7.779 critical value. In conclusion we accept H_0 on $1 - \alpha = 0.9$ degree.

Exercise

```
x = 0:4;
freq = [12 32 25 21 10];
prob = poisspdf(0:3,2)
prob = [prob,1-sum(prob)]
expected = prob*sum(freq)
[h,p,stats] = chi2gof(x,"Expected",expected,"Frequency",freq,"Alpha",0.1)
```

Exercise 1.7

H_0: X has a standard normal distribution.
Let Φ denote the standard normal distribution function. From the table of the standard normal distribution:

$$\Phi(0) = 0.5, \qquad \Phi(0.5) = 0.6915, \qquad \Phi(1) = 0.8413,$$
$$\Phi(1.5) = 0.9332, \Phi(2) = 0.9772.$$

We choose the following divider points:

$$-2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, 2$$

If H_0 is true, the probability of the intervals are the following:

$$p_1 = \Phi(-2) = 0.0228, \qquad p_2 = \Phi(-1.5) - \Phi(-2) = 0.0440,$$
$$p_3 = \Phi(-1) - \Phi(-1.5) = 0.0919, \qquad p_4 = \Phi(-0.5) - \Phi(-1) = 0.1498,$$
$$p_5 = \Phi(0) - \Phi(-0.5) = 0.1915, \qquad p_6 = \Phi(0.5) - \Phi(0) = 0.1915,$$
$$p_7 = \Phi(1) - \Phi(0.5) = 0.1498, \qquad p_8 = \Phi(1.5) - \Phi(1) = 0.0919,$$
$$p_9 = \Phi(2) - \Phi(1.5) = 0.0440, \qquad p_{10} = 1 - \Phi(2) = 0.0228.$$

Exercise 1.7

After $N = 1000$ observations on X the frequency of a point is falling into a certain interval is the following:

$$k_1 = 20, k_2 = 45, k_3 = 101, k_4 = 132, k_5 = 224,$$
$$k_6 = 190, k_7 = 156, k_8 = 87, k_9 = 41, k_{10} = 4.$$

The χ^2 statistics is:

$$\chi^2 = \frac{(20-22.8)^2}{22.8} + \frac{(45-44)^2}{44} + \frac{(101-91.9)^2}{91.9} + \frac{(132-149.8)^2}{149.8} + \frac{(224-191.5)^2}{191.5} + \frac{(190-191.5)^2}{191.5} + \frac{(156-149.8)^2}{149.8} + \frac{(87-91.9)^2}{91.9} + \frac{(41-44)^2}{44} + \frac{(4-22.8)^2}{22.8} =$$
$$= 0.344 + 0.023 + 0.901 + 2.115 + 5.516 + 0.012 + 0.257 + 0.261 + 0.205 + 15.502 = 25.136$$

We reject H_0 on every degree, since based on the χ_9^2 table at the 0.95 degree the critical value is 16.92, at the 0.99 degree the critical value is 21.67, at the 0.995 degree the critical value is 23.59. Our statistics value is greater than all of the above.

Exercise

```
freq = [20 45 101 132 224 190 156 87 41 4];
edges = [-2 -1.5 -1 -0.5 0 0.5 1 1.5 2];
expected = [normcdf(-2),normcdf(-1.5:0.5:2)-normcdf(-2:0.5:1.5),1-normcdf(2)]*sum(freq)
[h,p,stats] = chi2gof(-2.25:0.5:2.25,"Expected",expected,"Frequency",freq)
```

Exercise 1.8

Are the hair and eyes colour independent? After observing 200 people we got the following table:

	Blond hair	Brown hair	Black hair	$\sum$
Blue eyes	42	28	3	73
Brown eyes	17	89	21	127
$\sum$	59	117	24	200

$$\chi^2 = \frac{(42-21.54)^2}{21.54} + \frac{(28-42.71)^2}{42.71} + \frac{(3-8.76)^2}{8.76} + \frac{(17-37.47)^2}{37.47} + \frac{(89-74.30)^2}{74.30} + \frac{(21-15.24)^2}{15.24} =$$
$$= 19.43 + 14.71 + 3.79 + 11.18 + 2.91 + 2.18 = 49.11.$$

The degree of freedom is $(2-1)(3-1) = 2$. Since in the table of χ_2^2 even the $\alpha = 0.001$ degree's critical value is 13.82, we reject H_0 at all possible degrees.

Exercise

```
function [h,p,X2] = chi2cont(x,alpha)
e = sum(x,2)*sum(x)/sum(x(:));
X2 = (x-e).^2./e;
X2 = sum(X2(:));
df = prod(size(x)-[1 1]);
p = 1-chi2cdf(X2,df);
h = double(p<=alpha);
```

```
cont = [[42;17],[28;89],[3;21]];
[h,p,x2] = chi2cont(cont,0.05)
```

Literature

- Ágnes Baran: Mathematics for engineers 1. (Laboratory slides)
- Sándor Baran: Probability theory and statistics
- Fazekas István: Valószínűségszámítás, Debreceni Egyetemi Kiadó, 2009
- Fazekas István: Bevezetés a matematikai statisztikába, Debreceni Egyetemi Kiadó, 2009
- Matlab examples: https://www.mathworks.com/help/examples.html